

---

# Early Stopping Rollout: Early Tokens Are Better For On-Policy Distillation

---

Anonymous Author(s)

Affiliation

Address

email

## Abstract

1 On-policy knowledge distillation scores every token of the student’s rollout with  
2 the teacher. We identify a subtlety that changes the picture: the teacher’s score  
3 at position  $t$  is conditioned on the student’s own previously generated tokens, so  
4 early in a rollout the teacher judges *how to approach the problem*, whereas later  
5 it can only judge *how to continue what the student has already written*. Given a  
6 sufficiently long student rollout, the teacher’s accuracy collapses toward the student  
7 baseline, indicating that late-position supervision is far weaker than early-position  
8 supervision. We therefore propose *Early Stopping Rollout* (ESR): restrict both the  
9 rollout and the reverse-KL loss to the first  $N$  response tokens. Across tasks, model  
10 families, LoRA and full fine-tuning, and a range of teacher scales, ESR matches  
11 or exceeds full-sequence distillation nearly everywhere, sometimes surpasses the  
12 teacher, trains far faster with far less memory, and avoids the collapse modes of  
13 full-sequence training. The obvious explanation—ESR as a proxy for high-KL or  
14 high-entropy tokens—fails: every scalar-saliency selector we tried underperforms  
15 ESR, and most underperform plain full-sequence. Position is an independent,  
16 load-bearing axis—it is where the planning tokens live, and a short reverse-KL loss  
17 there is enough to improve the entire response.

## 18 1 Introduction

19 On-policy distillation (OPD) has emerged as a dominant paradigm for model distillation in industrial  
20 practice. The student generates its own rollouts  $\tau$ , which are then scored by the teacher: at each token,  
21 the teacher’s probability  $\pi_{teacher}(\tau_{student} \mid x_{prompt})$  serves as the soft target for the student [Agarwal  
22 et al., 2024, Gu et al., 2024]. Viewed through an RL lens, this is equivalent to using the teacher as a  
23 dense, token-level reward model that judges the student’s own behavior on a given prompt. The final  
24 objective is then a simple *uniform average* of the per-token reverse-KL across the trajectory.

25 This uniform averaging implicitly assumes that the teacher’s signal is equally reliable at every position  
26 in the trajectory. We argue that this assumption is unfounded, and that it breaks down precisely  
27 because of the autoregressive structure of LLMs. At the first token, the teacher’s score is conditioned  
28 only on the prompt:  $\pi_{teacher}(\tau_{student}^{t=1} \mid x_{prompt})$  - indeed what we expect the teacher to score on. At  
29 a later position  $m$ , however, it becomes conditioned also on the student’s own previously generated  
30 tokens:  $\pi_{teacher}(\tau_{student}^{t=m} \mid x_{prompt}, \tau_{student}^{t=1:m})$ . This conditioning context is off-policy to the teacher  
31 model and may even be out-of-distribution (OOD), and grows more so with position.

32 To probe how severe this drift is, we run a preliminary experiment that measures how the teacher  
33 behaves when continuing from an early-stopped student rollout. As shown in Figure 1, the teacher’s  
34 quality collapses toward the student’s within roughly 300 tokens. If the teacher’s late-position signal  
35 is unreliable, the natural question is whether early tokens alone are systematically more informative  
36 — and whether training on them is sufficient.

Motivated by this, we propose Early Stopping Rollout(ESR): restrict the student rollout to its first  $N$  tokens (e.g., 50 or 100) and compute the distillation loss only on this early window. The change is a single line in any on-policy distillation loop. Despite its simplicity, ESRconsistently outperforms full-rollout OPD across tasks (math, code, function calling), training regimes (LoRA, full fine-tuning), model scales (students 1.5B–7B, teachers 1.7B–14B), and model families (Qwen, Gemma), while reducing wall-clock cost by up to  $24\times$  and peak training memory by up to  $4\times$ . We further observe that full-rollout OPD frequently destabilizes and collapses during training, whereas ESRdoes not collapse in any of our settings. Perhaps most surprisingly, although the student is normally upper-bounded by the teacher’s performance, ESR-trained students often *exceed* the teacher.

Such a small change yields such large gains probe us to investigate deeper. We have verified that later position tokens suffer from signal quality issues. But we want to further validate whether the effectiveness of earlier tokens is intrinsically caused by better KL and entropy signals, because we also find that early tokens generally have larger KL and entropy. We test selectors that pick tokens by high KL, high entropy or combined regardless of position; all underperform ESR, with most underperforming full-rollout. Therefore, although we don’t exclude KL and entropy as auxillary factors, we confirm that the position is a an independent factor rather a saliency proxy. The late position tokens have signal quality issues in themselves. Furthermore, we identified two other mechanisms that may be working together: (i) *Strategy learning*: we find content-wise the early tokens tend to encode strategies and plans, while later tokens tend to focus on specific executions. It may be better to learn just the planning part from the teacher and leave the execution to its own way. (ii) *Convergence Cascade*: after training on the early window, KL divergence on the *untrained* late tokens also drops by 30–40%; the KL divergence does not have to see late positions to repair them.

These mechanisms explain why ESRcan beat full-rollout. They do not explain why ESRsometimes even *exceeds* the teacher — a stronger phenomenon we observe on most of our main experiment Table 1. We trace this to the mode-seeking nature of reverse-KL: the student commits to a subset of supported teacher modes instead of chasing all of them. Our ESRloss helps it steer toward a non-dominant but correct mode rather than the teacher’s argmax. Empirically, we find ESR-trained students commit harder to their own choices than full-rollout students yet agree *less* with the teacher’s argmax, and produce trajectories  $\sim 3\times$  shorter than full-rollout and  $\sim 2\times$  shorter than the base student (§5.3). It learns the teacher’s strategy but preserves its succinct style, leading to better compound behaviors. This finding showcases a potential path of superceding the teacher model in distillation that worth further investigations.

Our contributions are summarized below:

1. **Early Stopping Rollout(ESR) outperforms full-rollout on-policy distillation.** A one-line change—restricting the rollout length to the first  $N$  response tokens—beats full-sequence distillation across tasks, model families, scales, and training regimes, while being dramatically more efficient and far more stable to train. It can even exceed the teacher’s own performance (Sections 3–4.3).
2. **A systematic investigation of why it works.** First, through a token-selection ablation, we show that the gain is not “high KL” or “high entropy” in disguise: position itself is a load-bearing axis. We also identify strategy learning, and convergence cascading to be potential factors. Furthermore, we explain why ESRoften surpasses the teacher: by exploiting the mode-seeking behavior of reverse-KL training, ESRpushes the student toward a better, non-dominant sub-mode (Section 5).

## 2 Motivation: Why Late-Position Supervision Is Unreliable

In on-policy distillation, the teacher model scores the student’s rollout at each position  $t$  via  $\pi_t(\cdot | x, y_{<t}^{\text{student}})$  and averages these scores uniformly across the rollout. The conditioning context  $y_{<t}^{\text{student}}$ , however, is whatever this particular student happened to write at this particular moment of training. As  $t$  grows it drifts away from any distribution the teacher itself was trained on—off-policy to the teacher and, for long enough or atypical enough rollouts, plausibly out-of-distribution.

To verify the intuition, we feed the teacher a  $k$ -token student-generated rollout on MATH-500 and letting it continue autoregressively, the teacher’s avg@4 ( $n=4$ ,  $t=0.7$ ) falls from its 65.30% unconditional baseline to 62.70% at  $k=100$  and 51.75% at  $k=300$ —essentially student-baseline

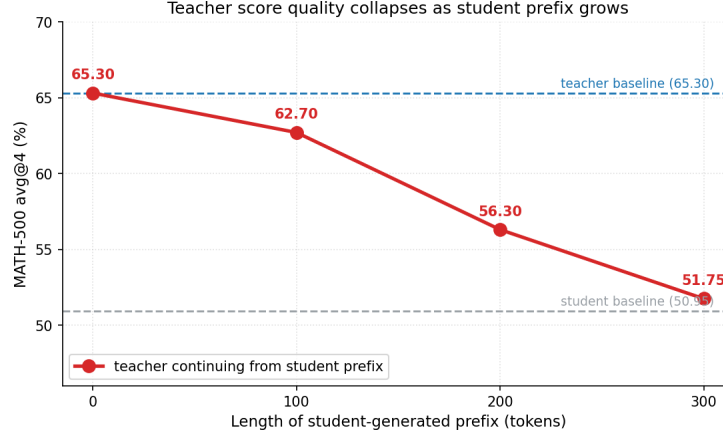


Figure 1: **The teacher loses accuracy quickly as the student-generated rollout grows in a few hundred of tokens.** MATH-500, avg@4 ( $n=4$ ,  $t=0.7$ ). Teacher = Qwen3-1.7B; student = Qwen2.5-Math-1.5B. After  $\sim 300$  student tokens the teacher has effectively been dragged down to student-baseline performance.

(Figure 1). Late-position teacher scores are not independent assessments of the prompt; it is a guess at how to continue a trajectory the student has already committed to.

The problem compounds during training because the student’s policy is moving. Early on, the student’s rollout is far from the teacher; late-position targets are fitted under a strongly off-policy prefix. As training progresses and earlier tokens approach the teacher’s distribution, the right late-position behavior changes too—continuations that were sensible above an off-policy prefix are no longer sensible above an on-policy one. But the late-position habits learned earlier do not cleanly erase: gradient descent gives no guarantee of unlearning, and in LoRA’s low-rank setting it updates make redirection compete for capacity that may already be spent. Full-rollout training thus chases a moving late-position target while carrying ballast from earlier ones. This insight expects training stability issues for full-rollout OPD – a prediction we later verify in §4.3

### 3 Method: Early Stopping Rollout

#### 3.1 Formulation

Let  $\pi_s$  denote the student and  $\pi_t$  the teacher. In standard on-policy reverse-KL distillation, the student generates a response  $\mathbf{y} = (y_1, \dots, y_T)$  conditioned on prompt  $x$ , and the loss is

$$\mathcal{L}_{\text{full}} = \mathbb{E}_{\mathbf{y} \sim \pi_s(\cdot|x)} \left[ \sum_{t=1}^T \text{KL}(\pi_s(\cdot | x, \mathbf{y}_{<t}) \parallel \pi_t(\cdot | x, \mathbf{y}_{<t})) \right]. \quad (1)$$

ESR (position cutoff  $N$ , with  $N \ll T$  in practice) truncates the student rollout to its first  $N$  tokens, and the loss is computed over exactly those tokens:

$$\mathcal{L}_{\text{ESR}}(N) = \mathbb{E}_{\mathbf{y} \sim \pi_s(\cdot|x), |\mathbf{y}| \leq N} \left[ \sum_{t=1}^{\min(N, |\mathbf{y}|)} \text{KL}(\pi_s(\cdot | x, \mathbf{y}_{<t}) \parallel \pi_t(\cdot | x, \mathbf{y}_{<t})) \right]. \quad (2)$$

If the student emits EOS before position  $N$ , the rollout terminates naturally. Everything else—generation temperature, LoRA target modules, optimizer, scorer—is unchanged from the standard on-policy KD loop.

Table 1: **Main results: ESR dominates Full Rollout across tasks and families.** Best score across training steps; metrics defined in §4 setup. ★: ESR surpasses the teacher reference. **Full Rollout** numbers mark the best step before degradation; ↓ flags cells where Full Rollout further decays with training (more details in Appendix C).

| Pair                           | Method              | MATH-500       |                | Coding         |                | BFCL           |
|--------------------------------|---------------------|----------------|----------------|----------------|----------------|----------------|
|                                |                     | avg@4          | pass@4         | HE             | HE+            | full_acc       |
| Qwen2.5-Math 1.5B → Qwen3 1.7B | Student baseline    | 50.95          | 72.80          | 31.10          | 26.20          | 2.70           |
|                                | Teacher reference   | 65.30          | 77.00          | 39.60          | 36.00          | 54.00          |
|                                | <b>Full Rollout</b> | 62.35↓         | 74.60↓         | 40.20↓         | 35.40↓         | 58.20          |
|                                | <b>ESR</b>          | <b>65.85 ★</b> | <b>79.80 ★</b> | <b>42.10 ★</b> | <b>36.00</b>   | <b>61.30 ★</b> |
| Gemma-2 2B → Gemma-3 4B        | Student baseline    | 13.45          | 28.20          | 23.78          | 20.10          | 73.17          |
|                                | Teacher reference   | 66.60          | 74.80          | 20.70          | 20.10          | 72.83          |
|                                | <b>Full Rollout</b> | 22.95          | 31.40          | 22.00↓         | 17.70↓         | 76.83          |
|                                | <b>ESR</b>          | <b>27.20</b>   | <b>39.40</b>   | <b>28.70 ★</b> | <b>24.40 ★</b> | <b>79.00 ★</b> |

## 4 Main Results: ESR Dominates Full-Sequence

### 4.1 Setup

**Models.** We evaluate across multiple student-teacher pairs. Our primary student is Qwen2.5-Math-1.5B [Qwen Team, 2024], paired with teachers of increasing scale: Qwen3-1.7B, Qwen3-4B, and Qwen3-8B [Qwen Team, 2025]. We also tested Qwen2.5-Math-7B with Qwen3-14B to test student model size generality. To test cross-family generality, we also evaluate with gemma-2-2b-it (student) and gemma-3-4b-it (teacher).

**Training.** All experiments use LoRA [Hu et al., 2022] ( $r=32$ ,  $\alpha=64$ , all linear layers,  $lr = 5 \times 10^{-5}$ ) with reverse KL divergence loss. Temperature 0.7 for on-policy generation. Each training step processes a batch of 16 problems with 1 rollout per problem ( $n_{\text{samples}}=1$ , batch size 16). We train for 200 steps on all tasks, and saving checkpoints every 50 steps. Training data are drawn from NuminaMath, CodeUltraFeedback, and glaive-function-calling-v2. Our method uses  $N=100$  unless otherwise specified.

**Evaluation.** MATH-500 [Hendrycks et al., 2021, Lightman et al., 2023] with  $n=4$  samples at temperature 0.7 (reporting avg@4), HumanEval/HumanEval+ [Chen et al., 2021, Austin et al., 2021, Liu et al., 2023] at temperature 0.0 (reporting pass@1), BFCL [Patil et al., 2025] with 600 problems reporting full accuracy: correct function name and arguments.

### 4.2 Results Across Families, Tasks, Regimes, and Scales

Table 1 presents the unified main results across tasks, model families, scales, and training regimes. We report the best performance achieved across all training steps for each configuration. But notice that full rollout distillation frequently collapse to much lower performance as it trains. We denote the sessions that collapses with a down arrows.

**Early Stopping Rollout beats full rollout across model families and tasks, and frequently surpasses teacher performance.** Across the cells of Table 1, ESR beats full rollout on essentially every cell, and in six cells ESR *surpasses the teacher* outright (Qwen math avg@4, pass@4; Qwen HumanEval and HumanEval+; Qwen BFCL full\_acc; Gemma coding on all four columns; Gemma BFCL full\_acc). The Gemma coding story is the sharpest: full rollout drives HumanEval pass@1 to 22.00, *below* the 23.78 student baseline and worse than the 20.70 teacher, while ESR recovers 31.10. Qwen-math full rollout is marked ↓ because its best number (62.35 avg@4) is a non-stable best step: under  $n=16$  it collapses via boxed-repetition to 46.3% by step 100 (Appendix C.6).

**Early Stopping Rollout matches or beats full rollout across full fine-tuning and LoRA settings.** The ESR story is not a LoRA artifact. Table 2 reports FullFT on MATH-500 for both Qwen and Gemma. On Qwen the gap narrows (**Full Rollout** 58.20 vs ESR 56.20 avg@4; both well below the

Table 2: **FullFT on MATH-500**. Best across training steps. **Bold**: ESR beats **Full Rollout**.

| Pair              | Method              | avg@4        | pass@4       |
|-------------------|---------------------|--------------|--------------|
| Qwen2.5-Math 1.5B | student/teacher     | 50.95/65.30  | 72.80/77.00  |
| → Qwen3 1.7B      | <b>Full Rollout</b> | 58.20        | 75.40        |
|                   | <b>ESR</b>          | 56.20        | 73.80        |
| Gemma-2 2B        | student/teacher     | 13.45/66.60  | 28.20/74.80  |
| → Gemma-3 4B      | <b>Full Rollout</b> | 13.90        | 25.00        |
|                   | <b>ESR</b>          | <b>26.65</b> | <b>40.40</b> |

Table 3: **Scaling teacher and student (LoRA) on MATH-500**.  $\downarrow$ : **Full Rollout** peak before degradation. **Bold**: ESR beats **Full Rollout**.

| Pair              | Method              | avg@4              | pass@4       |
|-------------------|---------------------|--------------------|--------------|
| Qwen2.5-Math 1.5B | student/teacher     | 50.95/77.95        | 72.80/86.40  |
| → Qwen3 4B        | <b>Full Rollout</b> | 67.45 $\downarrow$ | 80.60        |
|                   | <b>ESR</b>          | <b>68.95</b>       | <b>81.00</b> |
| Qwen2.5-Math 1.5B | student/teacher     | 50.95/77.70        | 72.80/87.80  |
| → Qwen3 8B        | <b>Full Rollout</b> | 66.75              | 81.60        |
|                   | <b>ESR</b>          | <b>67.85</b>       | <b>82.20</b> |
| Qwen2.5-Math 7B   | student/teacher     | 53.60/76.15        | 75.00/83.20  |
| → Qwen3 14B       | <b>Full Rollout</b> | 68.85 $\downarrow$ | 80.00        |
|                   | <b>ESR</b>          | <b>68.95</b>       | <b>81.20</b> |

LoRA-ESR best of 65.85). On Gemma ESR still dominates: ESR beats **Full Rollout** by +12.75pp avg@4 and +15.40pp pass@4. ESR is the safer choice in both parameter regimes.

**Early Stopping Rollout beats full rollout across model sizes.** The advantage persists as both teacher and student grow. Table 3 reports three scaling regimes. With the Qwen3-4B teacher, **Full Rollout** peaks at step 50 (67.45) and then *collapses* via boxed-repetition to 55.05 by step 200 — a −12.4pp drop across training — while ESR stays stable and climbs to 68.95. Against the 8B teacher, ESR still edges **Full Rollout** (67.85 vs 66.75); **Full Rollout** was only run at step 50, so no trajectory is available. The pattern persists at much larger scale: with a Qwen2.5-Math-7B student and Qwen3-14B teacher, **Full Rollout** again peaks at step 50 (68.85) and drifts to 62.40 by step 200, while ESR holds across all checkpoints.

### 4.3 Method Stability

**Early Stopping Rollouts are more robust than full rollout in training.** Across our (pair  $\times$  task  $\times$  regime) matrix, full-rollout distillation degrades with training in seven cells; ESR degrades nowhere. Appendix Table 14 reports each collapsing cell as a (peak  $\rightarrow$  lowest) pair with the corresponding step indices. The headline failures are: Qwen-LoRA coding falls from 40.2 at s50 to 26.8 by s350 ( $\Delta = -13.4$ pp); Qwen-LoRA math under  $n_{\text{samples}}=16$  collapses via boxed-repetition from 65.60 at s50 to 43.80 by s200 ( $\Delta = -21.8$ pp, ending below the 50.95% student baseline); Qwen-LoRA math with the Qwen3-4B teacher peaks at 67.45 at s50 and drifts to 54.45 by s100 ( $\Delta = -13.0$ pp); and Gemma-LoRA coding peaks at 22.0 at s50 and falls to 11.6 ( $\Delta = -10.4$ pp), more than 12pp below the student baseline. Three of seven full-rollout collapses end below the student baseline. ESR on the same cells stays within  $\pm 1$ pp of its peak across training (Tables 9, 11).

**Early Stopping Rollouts are not sensitive to the choice of  $N$ .** Figure 2 sweeps  $N$  on MATH-500 and reveals a robust sweet spot: all rollout cutoffs from  $N=50$  upward converge to  $\sim 66\%$  avg@4, outperforming full-sequence distillation (62.35%) by 3–4pp. The method is not sensitive to the exact choice of  $N$  within this plateau, so no per-task tuning is required.

### 4.4 Efficiency

Table 4 shows that ESR achieves a  $24\times$  wall-clock speedup and reduces peak training memory by  $\sim 4\times$ . The dominant cost in full-seq is autoregressive generation (180 s/step for sequences averaging  $\sim 1000$  tokens); ESR generates only  $N=100$  tokens (5 s/step). ESR also does not require a vLLM/SGLang inference server—100 tokens can be generated via standard HF generate. Full-seq with an 8B teacher OOMs on a 48 GB A6000; ESR runs comfortably, unlocking larger teachers on smaller GPUs.

## 5 Why Does ESR Work?

The result that a one-line change matches or exceeds full-sequence distillation — at a fraction of the compute, with stability everywhere the full-sequence baseline degrades — is surprising enough

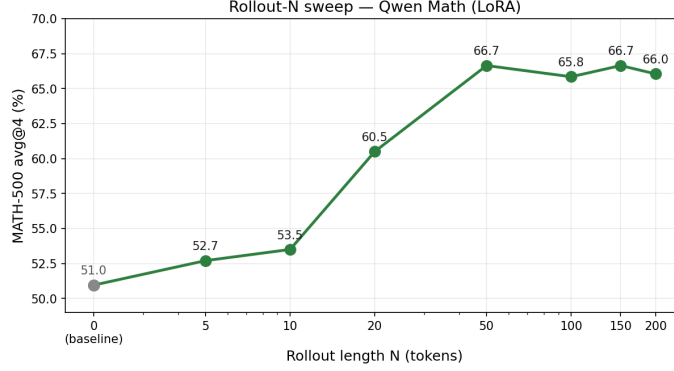


Figure 2: **Rollout Length  $N$  Sweep on MATH-500.** LoRA, Qwen2.5-Math-1.5B  $\rightarrow$  Qwen3-1.7B,  $n=1$ , 3,200 problems; best avg@4 across training steps. Full-seq and the undistilled baseline are shown as horizontal references. Performance saturates for  $N \in [50, 200]$  and all points in this range beat full-seq.

| Phase                      | Per-step wall time |                 |                              | Peak GPU memory   |                 |                               |
|----------------------------|--------------------|-----------------|------------------------------|-------------------|-----------------|-------------------------------|
|                            | ESR<br>(100 tok.)  | Full<br>rollout | Speedup                      | ESR<br>(100 tok.) | Full<br>rollout | Savings                       |
| Teacher scoring            | 1 s                | 7 s             | 7 $\times$                   | 7.3 GB            | 14.9 GB         | 2.0 $\times$                  |
| Student generation         | 5 s                | 180 s           | 36 $\times$                  | 7.2 GB            | 8.9 GB          | 1.2 $\times$                  |
| Student training           | 2 s                | 7 s             | 3.5 $\times$                 | 9.6 GB            | 39.5 GB         | 4.1 $\times$                  |
| <b>Total (single step)</b> | <b>8 s</b>         | <b>194 s</b>    | <b>24<math>\times</math></b> | <b>24.1 GB</b>    | <b>63.3 GB</b>  | <b>2.6<math>\times</math></b> |
| <b>Total (200 steps)</b>   | <b>27 min</b>      | <b>10.8 hr</b>  | <b>24<math>\times</math></b> |                   |                 |                               |

Table 4: **Training efficiency.** Single A6000 (48 GB), bs=16,  $n=1$ , teacher Qwen3-1.7B. Per-phase wall time and peak GPU memory share the same phases. Phases run sequentially; the memory “Total” columns span both rows because they describe the co-resident footprint, i.e., the sum of per-phase peaks if teacher and student stay resident simultaneously instead of being swapped in/out.

178 that it demands a deeper mechanism. We structure this section around three questions: (§5.1) Is  
 179 the early-position effect simply a proxy for high-KL or high-entropy tokens? (§5.2) If not, what  
 180 additional structural reasons make the early window privileged? (§5.3) Why does ESR sometimes  
 181 *exceed* the teacher?

## 182 5.1 Is the Effectiveness of the Early Position Caused by KL and Entropy?

183 Figure 3 shows that early positions are simultaneously high-KL and high-entropy, so a natural  
 184 alternative to positional selection is *information-theoretic* selection: pick the  $k$  tokens with the highest  
 185 KL divergence or highest student/teacher entropy, regardless of position. We tested every such  
 186 selector (Table 5): all underperform ESR, and most underperform full-sequence. For each strategy in  
 187 Table 5 we keep the full 400-token on-policy rollout, compute the selector, and apply the reverse-KL  
 188 loss only to the selected tokens.

189 Two findings fall out of Table 5. (i) Top-RKL=100, the selector that captures the most raw KL signal,  
 190 is the *worst* (53.35%) — worse than no distillation. Increasing  $k$  to 200 only partly recovers. (ii) Every  
 191 KL/entropy/combined selector underperforms full-sequence (62.35%), which in turn underperforms  
 192 ESR (65.85%). These selectors all concentrate on the contaminated late region where the teacher  
 193 is scoring its own continuation of the student’s previously generated tokens; the benefit of the early  
 194 region is therefore *structural*, not reducible to any scalar per-token statistic.

195 **Takeaway.** ESR is not mainly caused by KL or entropy. Position is an independent, load-bearing  
 196 axis — not “high-KL in disguise” and not “high-entropy in disguise”. Whatever makes the early  
 197 window privileged is structural; we identify two such reasons next.



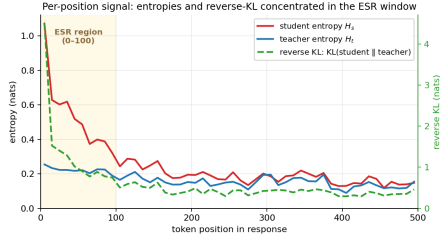


Figure 3: **Early positions carry all three — student entropy, teacher entropy, and reverse-KL — simultaneously** (Qwen, MATH-500). Natural hypothesis: ESR works because those regions carry the most signal (high KL) at the most uncertain points (high entropy). If that were the mechanism, selecting tokens by high-KL or high-entropy alone should recover the gain. Table 5 shows it does not.

| Selection method                       | $k$ | Best avg@4   |
|----------------------------------------|-----|--------------|
| <b>ESR-100 (ours)</b>                  | 100 | <b>65.85</b> |
| Full-seq (all tokens)                  | —   | 62.35        |
| Top-RKL                                | 100 | 53.35        |
| Top-RKL                                | 200 | 56.15        |
| Top- $H_t$ (teacher entropy)           | 100 | 63.30        |
| Top- $H_s$ (student entropy)           | 100 | 62.70        |
| Top- $H_s$                             | 200 | 53.20        |
| RKL · $H_s$                            | 100 | 56.90        |
| $H_t \cdot H_s$ (product)              | 100 | 55.35        |
| RKL · $H_t \cdot H_s$ (triple product) | 100 | 57.90        |

Table 5: **Token selection strategies on MATH-500.** Qwen2.5-Math-1.5B  $\rightarrow$  Qwen3-1.7B, LoRA,  $n_{\text{samples}}=1$ , best avg@4 across training steps. All selectors operate on full-length rollouts; ESR is the only one that additionally truncates the rollout. Baseline (no distillation) 50.95%; full-seq 62.35%.  $H_s$ : student entropy;  $H_t$ : teacher entropy; RKL: reverse KL.

## 198 5.2 Strategy Learning and Convergence Cascade

199 Section 5.1 rules out scalar per-token saliency. Beyond the late-position contamination already  
200 established in §2, we identify two further structural mechanisms that jointly explain why training on  
201 the early window is so effective.

202 **Strategy learning — the early window is where planning tokens live.** The early-position KL  
203 spike concentrates on planning / structural tokens: the highest-KL tokens in our corpus are “To” (KL  
204 6.90,  $n=8806$ ), “First” (9.98), “Let”, “The”, “Therefore” (7.95), “Thus”, along with format-opening  
205 tokens  $\backslash$  [,  $\backslash$  (, and code-fence ‘ ‘ ‘ (see Table 17). These tokens *select* the reasoning trajectory —  
206 different openings commit the student to algebra-with-variables, factoring, or numeric-estimation  
207 paths on the same problem. The teacher’s conditional distribution at these tokens is genuinely  
208 multi-modal (entropy 1.07), so reverse-KL has room to move the student to a single teacher-supported  
209 mode. After the plan is committed, the remainder is execution: the teacher is near-deterministic  
210 (entropy 0.14) and there is little signal to extract. *Teach the plan; the execution follows* — and no  
211 scalar saliency can recover the early-window region because what matters is where in the rollout a  
212 token lives, not how surprising it is locally.

213 **Convergence cascade — training the early window lowers KL everywhere.** A natural worry  
214 is that ESR helps only inside the trained window. The opposite holds. After ESR-200 training,  
215 per-position KL between distilled student and teacher drops by 50–60% inside  $[0, N]$  and by 30–40%  
216 outside it (Figure 4). Comparing distilled vs. base responses on the same prompts, late tokens —  
217 which receive zero direct supervision — change as much as early tokens in surface content (Jaccard to  
218 baseline = 0.251 for both halves). Autoregressive generation explains the cascade: once the student  
219 has committed to the right plan in the first  $N$  tokens, the remaining rollout is a lower-KL, more  
220 on-policy completion for the teacher, so no gradient is needed there. The loss does not have to *see*  
221 late positions to align them.

## 222 5.3 Why Does ESR Sometimes Exceed the Teacher?

223 ESR exceeds the teacher on several cells (Qwen math; Qwen and Gemma function calling; Gemma  
224 coding, where the teacher is in fact weaker than the student). We offer a hypothesis. Reverse KL  
225  $\text{KL}(\pi_s \parallel \pi_t)$  is mode-seeking: it penalizes the student for putting mass on tokens the teacher does not  
226 support, but not for concentrating mass on a single supported token. When the teacher is multi-modal  
227 at the planning positions—one mode being a verbose, wasteful plan and another a concise correct  
228 plan—a short early-window loss can force the student to commit to the smaller, correct mode rather  
229 than averaging (Figure 5). On function-calling this effect is pronounced: ESR-100 lifts Qwen BFCL  
230 full\_acc from 54.0% (teacher) to 61.3%, and the chained Qwen3-4B $\rightarrow$ 8B configuration raises it to

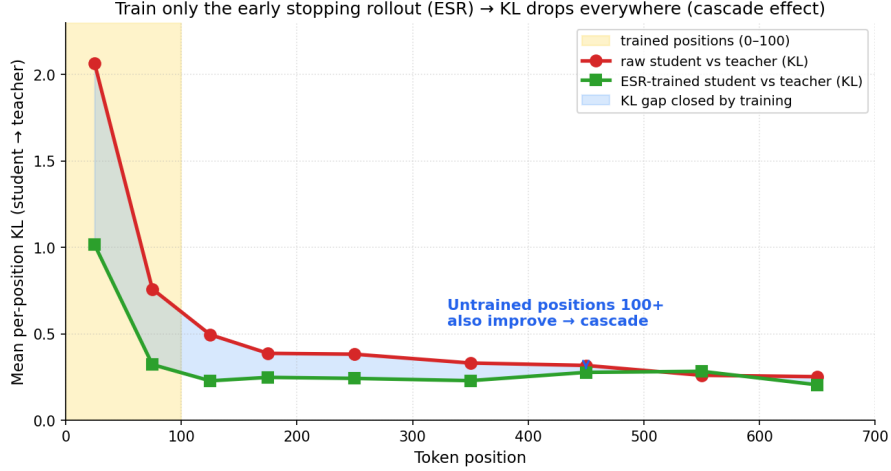


Figure 4: **The convergence cascade.** Per-position KL between distilled student and teacher, before vs. after ESR-200 training. Yellow band: positions  $[0, 200]$  that actually receive training loss. Blue band: the KL gap closed by training. Positions  $200+$  — which see no direct training signal — drop to the same KL as the trained region, confirming that alignment on the early window cascades through the autoregressive rollout.

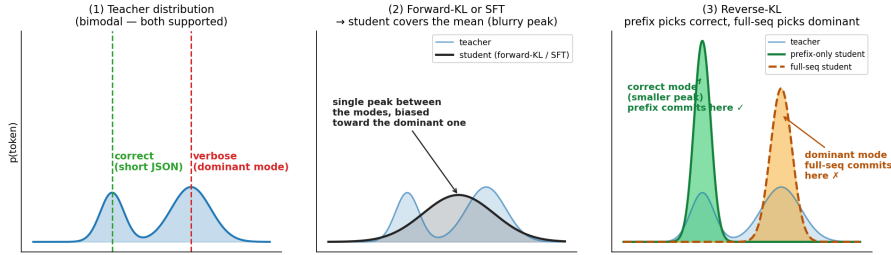


Figure 5: **Mode-seeking on the planning region.** Schematic of reverse-KL behaviour at a multi-modal planning position. The teacher supports two modes (e.g. a verbose plan and a concise correct plan); reverse KL penalises student mass outside the support but not concentration within it. A short early-window loss can collapse the student onto the better supported mode, allowing the student to exceed the teacher’s average behaviour. Full-sequence training reverts the student toward the averaged teacher across late positions, undoing the concentration.

231 76.8% against a 73.0% teacher. Full-sequence training cannot achieve this because its late-position  
 232 loss pulls the student toward the average teacher behaviour across all supported modes, undoing the  
 233 concentration.

234 **Behavioral signature.** The hypothesis predicts that ESR-trained students should (a) produce  
 235 shorter, less-hedging rollouts than the teacher, and (b) commit hard, but on *non-argmax* teacher  
 236 modes. Table 6 confirms both. ESR-100 rollouts are shorter than the base student, the teacher, *and*  
 237 full-seq across the entire length distribution — even the top-decile ( $\sim 800$  tokens) is below the base  
 238 student’s median ( $\sim 860$ ). On the top-10% highest-KL positions ( $n=110,894$ ), ESR raises student  
 239 top-1 probability to 0.79 (vs 0.77 full-seq, 0.71 base) while *lowering* agreement with the teacher’s  
 240 argmax to 41.9% (vs 45.7% full-seq) and raising the rate at which the student picks tokens in the  
 241 teacher’s top-2 to top-5 to 47.4% (vs 44.6%). ESR commits harder, but on different teacher-supported  
 242 tokens; full-seq commits toward the teacher’s argmax and inherits the teacher’s mistakes on those  
 243 positions.

244 This is the mode-seeking mechanism in numbers. When one of those teacher-supported alternatives  
 245 happens to be the correct mode, ESR exceeds the teacher; full-seq cannot, because supervising the  
 246 late tail pulls the student toward the teacher’s argmax and the teacher’s mistakes along with it. A full  
 247 proof of this hypothesis is out of scope; we note only that it is the natural reverse-KL-mode-seeking



| Model                 | Response length<br>(tokens) |               |               |        |       | Student commitment<br>& mode alignment |                 |                   |                 |
|-----------------------|-----------------------------|---------------|---------------|--------|-------|----------------------------------------|-----------------|-------------------|-----------------|
|                       | 10th<br>ptile               | 50th<br>ptile | 90th<br>ptile | Mean   | Ratio | Top-1<br>prob                          | = tch.<br>top-1 | ∈ tch.<br>top 2–5 | ∉ tch.<br>top-5 |
| Base student          | ~400                        | ~860          | ~1,500        | ~990   | 2.2×  | 0.71                                   | 28.6%           | 59.8%             | 11.5%           |
| Teacher (Q3-1.7B)     | ~700                        | ~1,150        | ~1,800        | ~1,210 | 2.6×  | —                                      | —               | —                 | —               |
| Full-seq              | ~1,190                      | ~1,530        | ~1,770        | ~1,480 | 3.2×  | 0.77                                   | <b>45.7%</b>    | 44.6%             | 9.7%            |
| <b>ESR-100 (ours)</b> | ~190                        | ~380          | ~800          | ~460   | 1.0×  | <b>0.79</b>                            | 41.9%           | <b>47.4%</b>      | 10.7%           |

Table 6: **Behavioral signature of ESR.** *Left:* Response length distribution on MATH-500 ( $n=2,000$  rollouts per model; tokens estimated as chars/3.5). ESR-100 is shorter than base, teacher, and full-seq across the whole distribution: even its long tail (90th ptile  $\sim 800$  tok) is shorter than the base student’s typical response (50th ptile  $\sim 860$  tok). **Ratio** = mean length relative to ESR. Full-seq mean is below its median because length is clipped at `max_new_tokens`. *Right:* Top-10% of token positions ranked by  $KL[\pi_s \parallel \pi_t]$  on 100 base-student trajectories ( $n=110,894$ ). “= tch. top-1”, “∈ tch. top 2–5”, and “∉ tch. top-5” are mutually exclusive and sum to 100% (teacher omitted because the comparison is student-vs-teacher). ESR-100 commits hardest (top-1 prob 0.79) but agrees *less* with the teacher’s argmax than full-seq, and picks teacher-supported alternatives more often.

248 story applied to the planning region, and that it is consistent with every surpass-teacher cell we  
249 observe.

## 250 6 Related Work

251 **Knowledge Distillation for Language Models.** Knowledge distillation [Hinton et al., 2015]  
252 transfers knowledge from a teacher to a smaller student via soft targets. Kim and Rush [2016]  
253 extended this to sequence models with word-level and sequence-level objectives. For autoregressive  
254 LLMs, the choice of divergence and data distribution is decisive. Gu et al. [2024] argued for  
255 reverse KL in generative LLM distillation, on the grounds that reverse KL avoids the student placing  
256 probability on low-support teacher regions; Agarwal et al. [2024] introduced Generalized Knowledge  
257 Distillation (GKD), which generates rollouts from the student (“on-policy”) and supervises with  
258 teacher logits—substantially outperforming supervised off-policy distillation on reasoning tasks. Ko  
259 et al. [2024] proposed skew KL and an adaptive off-policy schedule; Wen et al. [2023] generalized  
260 sequence-level KD to arbitrary  $f$ -divergences; Wu et al. [2025] revisited the mode- vs. mean-seeking  
261 interpretation of reverse and forward KL in the LLM setting and showed that both converge under  
262 sufficient training but differ in which distribution regions they first concentrate on. Xu et al. [2025]  
263 introduced Speculative Knowledge Distillation, which interleaves teacher-student sampling in the  
264 style of speculative decoding and filters tokens the teacher would not have produced. Our work builds  
265 directly on the on-policy reverse-KL paradigm of Gu et al. [2024] and Agarwal et al. [2024], but  
266 asks a different question: holding the divergence and data distribution fixed, *which token positions*  
267 *along the student’s rollout carry useful signal?* Our answer—only the early window (Early Stopping  
268 Rollout)—is orthogonal to the divergence-design axis these works explore.

269 **Token-Level Importance in Distillation and Reasoning.** A growing body of work demonstrates  
270 that not all tokens contribute equally to learning. Wang et al. [2025] showed that only  $\sim 20\%$  of chain-  
271 of-thought tokens have high entropy, and these “forking tokens” steer reasoning paths—applying  
272 RL gradients to only these tokens matches full-gradient performance. In the RL fine-tuning setting,  
273 Vassoyan et al. [2025] demonstrated that uniform KL penalties across all token positions prevent  
274 exploration on “critical tokens” where the pretrained model is most uncertain, and proposed entropy-  
275 weighted KL relaxation that allows the model to deviate from its prior precisely where it needs to  
276 learn. Li et al. [2025] discovered a “preplan-and-anchor” mechanism in LLM attention patterns,  
277 where models first perform long-range contextual reference before producing semantic anchor tokens,  
278 enabling targeted credit assignment. Singh and Hakkani-Tür [2026] proposed greedy pruning of  
279 reasoning chains and showed that LLMs internally encode token-level functional importance for  
280 answer generation.

281 In the distillation setting specifically, several concurrent works have explored token selection. Huang  
282 et al. [2025] proposed SelecTKD, a propose-and-verify framework where the teacher verifies student-

proposed tokens, applying full loss to accepted tokens and masking rejected ones. Xie et al. [2026] introduced AdaKD, which adaptively adjusts token-level temperature based on learning stability. Tavor et al. [2026] systematically disentangled selective KD along position, class, and sample axes, proposing student-entropy-guided position selection (SE-KD) that reduces wall time by 70% without sacrificing accuracy. Kim and Baek [2026] combined entropy-based token selection with preference ranking in a dual distillation framework.

These works collectively establish that not all tokens deserve equal gradient weight, and most of them pick the “important” tokens via a scalar per-token criterion—entropy, KL, difficulty, or a verification score. Our Section 5.1 ablation directly contradicts the premise that such scalars are sufficient: selecting the same number of tokens by top-KL, top-entropy (student, teacher, AND, OR, triple-product), soft-weighted joint entropy, or top- $\hat{K}$ -renorm RKL (a vocabulary-level renormalization of RKL proposed for stabilizing distillation; exact citation to be added in camera-ready)<sup>1</sup> all underperform ESR and most underperform plain full-sequence training. Position is the load-bearing axis; scalar token saliencies are not. This also resolves a tension with Wang et al. [2025], who find that RL gradients concentrated on high-entropy “forking tokens” suffice: in on-policy *distillation*, the high-entropy tokens are disproportionately in the uncontaminated early window, so their result is consistent with ours once this early-window concentration of entropy is accounted for.

**Training Stability and Sequence-Level Issues.** Full-sequence on-policy distillation is known to suffer from instability. Holtzman et al. [2020] identified that likelihood-based training can produce bland, repetitive text—a form of neural text degeneration. Shumailov et al. [2024] demonstrated that iteratively training generative models on their own outputs leads to model collapse, where distribution tails disappear. In on-policy KD specifically, Jang et al. [2026] proposed Veto, a geometric bridge in logit space that suppresses harmful gradients on low-confidence tokens, directly addressing the instability we also observe. Our approach is complementary: rather than modifying the target distribution, we restrict the loss scope to early positions where teacher-student agreement is naturally highest, achieving stability through a simpler mechanism.

Several works have studied how sequence length affects distillation quality. Chen et al. [2025] showed that training on only the first 50% of tokens in supervised reasoning distillation retains ~91% of full-sequence performance, establishing a computation-quality tradeoff. Yan et al. [2026] addressed exposure bias and capacity misalignment in long chain-of-thought distillation through temperature-scheduled learning. Our work extends the truncation insight to the on-policy setting: we demonstrate that positional loss masking on student-generated sequences achieves equal or better performance than full-sequence loss, while providing the additional benefit of eliminating degenerate repetition patterns that are unique to on-policy training.

**Curriculum Learning and Sequence Scheduling.** Curriculum learning [Bengio et al., 2009] improves training by gradually increasing task difficulty. Bengio et al. [2015] applied this principle to sequence models through scheduled sampling, transitioning from teacher-forced to free-running generation during training. In the LLM pretraining setting, Li et al. [2022] showed that long sequences at training start cause extreme gradient variance, and sequence length warmup stabilizes training while enabling larger batch sizes. Pouransari et al. [2024] extended this with variable sequence length curriculum for LLM pretraining, achieving up to  $3\times$  faster convergence. Our progressive position schedule—linearly increasing the position limit from 1 to 200 over training—applies the curriculum principle to the distillation loss scope rather than to the input sequence length, and represents a novel form of curriculum specific to knowledge distillation.

**Reasoning Distillation.** Chain-of-thought prompting [Wei et al., 2022] made intermediate reasoning the unit of transfer for distillation, and the DeepSeek-R1 release [DeepSeek-AI, 2025] established that SFT-distilling a strong reasoner into a smaller model is often competitive with direct RL on the small model. Our experiments include three reasoning tasks (MATH-500, HumanEval/MBPP, BFCL) and a 1.5B-math-student  $\rightarrow$  8B-teacher sweep; ESR surpasses the teacher on several of these cells (Qwen math, function-calling), indicating that reasoning-distillation gains beyond the teacher are possible when the mode-seeking reverse-KL loss is restricted to the planning region. This is a complementary perspective to the DeepSeek-R1 recipe, which uses fully-sequenced SFT: our analysis explains why the on-policy, early-stopped-rollout regime is both more efficient and more stable for

<sup>1</sup>TODO: resolve and cite the top- $\hat{K}$ -renorm RKL method (Fu et al., 2025).

reasoning-style chain-of-thought distillation, and why full-sequence on-policy distillation—which in principle should be the strongest setting—frequently degrades across training on coding and math.

## 7 Conclusion

In on-policy knowledge distillation, the teacher’s per-token supervision quality is a strong, monotonic function of position: early tokens carry strong, independent signal about reasoning strategy; late tokens, scored on the student’s own contaminated previous output, contribute weak, dependent signal that is actively harmful when trained on. A one-line intervention — restrict the loss and the rollout to the first  $N$  response tokens — yields four simultaneous wins: (i) **better**, matching or exceeding full-sequence distillation on every task/family/regime we test and surpassing the teacher in nine cells; (ii) **faster**, 20–35 $\times$  wall-clock with 4 $\times$  less GPU memory; (iii) **stabler**, eliminating HumanEval collapse, BFCL parse-rate collapse, and the MATH-500 boxed-repetition failure mode; and (iv) **structural**, in that selecting the same number of tokens by any KL or entropy saliency fails — position is an independent axis, not a proxy for scalar surprise. Teaching the student how to begin is sufficient to improve the entire response.

**Limitations.** Our experiments are conducted on small open-source models with a limited data budget; whether the ESR story holds at industrial scale (much larger students, teachers, and training corpora) remains unclear. We also do not test multi-modality or long-horizon tasks, both of which may exhibit different positional signal-quality patterns.

## References

- Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos, Matthieu Geist, and Olivier Bachem. On-policy distillation of language models: Learning from self-generated mistakes. In *International Conference on Learning Representations*, 2024.
- Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, et al. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*, 2021.
- Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. Scheduled sampling for sequence prediction with recurrent neural networks. In *Advances in Neural Information Processing Systems*, 2015.
- Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. Curriculum learning. In *Proceedings of the 26th Annual International Conference on Machine Learning*, pages 41–48, 2009.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating large language models trained on code. *arXiv preprint arXiv:2107.03374*, 2021.
- Wei-Rui Chen, Vignesh Kothapalli, Ata Fatahibaarzi, Hejian Sang, Shao Tang, Qingquan Song, Zhipeng Wang, and Muhammad Abdul-Mageed. Distilling the essence: Efficient reasoning distillation via sequence truncation. *arXiv preprint arXiv:2512.21002*, 2025.
- DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. MiniLLM: Knowledge distillation of large language models. In *International Conference on Learning Representations*, 2024.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Advances in Neural Information Processing Systems*, 2021.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. In *NIPS Deep Learning and Representation Learning Workshop*, 2015.

382 Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text  
383 degeneration. In *International Conference on Learning Representations*, 2020.

384 Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang,  
385 and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International  
386 Conference on Learning Representations*, 2022.

387 Haiduo Huang, Jiangcheng Song, Yadong Zhang, and Pengju Ren. SelecTKD: Selective token-  
388 weighted knowledge distillation for LLMs. *arXiv preprint arXiv:2510.24021*, 2025.

389 Ijun Jang, Jewon Yeom, Juan Yeo, Hyunggu Lim, and Taesup Kim. Stable on-policy distillation  
390 through adaptive target reformulation. *arXiv preprint arXiv:2601.07155*, 2026.

391 Minsang Kim and Seung Jun Baek. Explain in your own words: Improving reasoning via token-  
392 selective dual knowledge distillation. *arXiv preprint arXiv:2603.13260*, 2026.

393 Yoon Kim and Alexander M. Rush. Sequence-level knowledge distillation. In *Proceedings of the  
394 2016 Conference on Empirical Methods in Natural Language Processing*, pages 1317–1327, 2016.

395 Jongwoo Ko, Sungnyun Kim, Tianyi Chen, and Se-Young Yun. DistiLLM: Towards streamlined  
396 distillation for large language models. In *International Conference on Machine Learning*, 2024.

397 Conglong Li, Minjia Zhang, and Yuxiong He. The stability-efficiency dilemma: Investigating  
398 sequence length warmup for training GPT models. In *Advances in Neural Information Processing  
399 Systems*, 2022.

400 Yang Li, Zhichen Dong, Yuhao Sun, Weixun Wang, Shaopan Xiong, Yijia Luo, Jiashun Liu, Han  
401 Lu, Jiamang Wang, Wenbo Su, Bo Zheng, and Junchi Yan. Attention illuminates LLM reason-  
402 ing: The preplan-and-anchor rhythm enables fine-grained policy optimization. *arXiv preprint  
403 arXiv:2510.13554*, 2025.

404 Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan  
405 Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. *arXiv preprint  
406 arXiv:2305.20050*, 2023.

407 Jiawei Liu, Chunqiu Steven Xia, Yuyao Wang, and Lingming Zhang. Is your code generated by  
408 ChatGPT really correct? rigorous evaluation of large language models for code generation. In  
409 *Advances in Neural Information Processing Systems*, 2023.

410 Shishir G. Patil, Huanzhi Mao, Fanjia Yan, Charlie Ji, Vishnu Suresh, Ion Stoica, and Joseph E.  
411 Gonzalez. The Berkeley function calling leaderboard (BFCL): From tool use to agentic evaluation  
412 of large language models. In *Proceedings of the 42nd International Conference on Machine  
413 Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 48371–48392, 2025.

414 Hadi Pouransari, Chun-Liang Li, Jen-Hao Rick Chang, Pavan Kumar Anasosalu Vasu, Cem Koc,  
415 Vaishaal Shankar, and Oncel Tuzel. Dataset decomposition: Faster LLM training with variable  
416 sequence length curriculum. In *Advances in Neural Information Processing Systems*, 2024.

417 Qwen Team. Qwen2.5-math technical report: Toward mathematical expert model via self-  
418 improvement. *arXiv preprint arXiv:2409.12122*, 2024.

419 Qwen Team. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

420 Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Nicolas Papernot, Ross Anderson, and Yarin Gal. AI  
421 models collapse when trained on recursively generated data. *Nature*, 631(8022):755–759, 2024.

422 Janvijay Singh and Dilek Hakkani-Tür. Do LLMs encode functional importance of reasoning tokens?  
423 *arXiv preprint arXiv:2601.03066*, 2026.

424 Almog Tavor, Itay Ebenspanger, Neil Cnaan, and Mor Geva. Rethinking selective knowledge  
425 distillation. *arXiv preprint arXiv:2602.01395*, 2026.

426 Jean Vassoyan, Nathanael Beau, and Roman Plaud. Ignore the KL penalty! boosting exploration  
427 on critical tokens to enhance RL fine-tuning. In *Findings of the Association for Computational  
428 Linguistics: NAACL 2025*, 2025.

- 429 Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen,  
430 Jianxin Yang, Zhenru Zhang, Yuqiong Liu, An Yang, Andrew Zhao, Yang Yue, Shiji Song, Bowen  
431 Yu, Gao Huang, and Junyang Lin. Beyond the 80/20 rule: High-entropy minority tokens drive  
432 effective reinforcement learning for LLM reasoning. In *Advances in Neural Information Processing  
433 Systems*, 2025.
- 434 Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc  
435 Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In  
436 *Advances in Neural Information Processing Systems*, 2022.
- 437 Yuqiao Wen, Zichao Li, Wenyu Du, and Lili Mou. f-divergence minimization for sequence-level  
438 knowledge distillation. In *Proceedings of the 61st Annual Meeting of the Association for Computa-  
439 tional Linguistics (Volume 1: Long Papers)*, pages 10817–10834, 2023.
- 440 Taiqiang Wu, Chaofan Tao, Jiahao Wang, Runming Yang, Zhe Zhao, and Ngai Wong. Rethinking  
441 Kullback-Leibler divergence in knowledge distillation for large language models. In *Proceedings  
442 of the International Conference on Computational Linguistics (COLING)*, 2025.
- 443 Xurong Xie, Zhucun Xue, Jiafu Wu, Jian Li, Yabiao Wang, Xiaobin Hu, Yong Liu, and Jiangn-  
444 ing Zhang. LLM-oriented token-adaptive knowledge distillation. In *Proceedings of the AAAI  
445 Conference on Artificial Intelligence*, 2026.
- 446 Wenda Xu, Rujun Han, Zifeng Wang, Long T. Le, Dhruv Madeka, Lei Li, William Yang Wang,  
447 Rishabh Agarwal, Chen-Yu Lee, and Tomas Pfister. Speculative knowledge distillation: Bridging  
448 the teacher-student gap through interleaved sampling. In *International Conference on Learning  
449 Representations*, 2025.
- 450 Shaotian Yan, Kaiyuan Liu, Chen Shen, Bing Wang, Sinan Fan, Jun Zhang, Yue Wu, Zheng Wang,  
451 and Jieping Ye. Distribution-aligned sequence distillation for superior long-CoT reasoning. *arXiv  
452 preprint arXiv:2601.09088*, 2026.

## 453 A Training Efficiency

Table 7: **Training efficiency, detailed breakdown.** Per-step wall-clock time on a single A6000 (48GB). Student: Qwen2.5-Math-1.5B, LoRA, bs=16.

| Teacher | Method           | Gen (s) | Score (s) | Train (s) | Total (s)         |
|---------|------------------|---------|-----------|-----------|-------------------|
| Q3-1.7B | Full-seq         | 100–731 | 3–10      | 5–12      | ~280              |
|         | Ours ( $N=100$ ) | ~5      | ~1        | ~2        | ~8                |
| Q3-4B   | Full-seq         | 97–733  | 3–9       | 6–10      | ~210              |
|         | Ours ( $N=100$ ) | ~5      | ~1        | ~2        | ~8                |
| Q3-8B   | Full-seq         | 96–230  | 7–11      | 1–6       | ~170 <sup>†</sup> |
|         | Ours ( $N=100$ ) | ~5      | ~1        | ~2        | ~8                |

<sup>†</sup> Frequent OOMs; 48GB insufficient for 8B teacher + vLLM + student.

## 454 B Additional Student-Teacher Pairs

Table 8: **Additional student-teacher pairs.** All use ESR-100, LoRA, 3,200 problems.

| Student    | Teacher      | MATH-500 avg@4 | BFCL full_acc | HumanEval pass@1 |
|------------|--------------|----------------|---------------|------------------|
| Qwen3-1.7B | — (baseline) | 69.20%         | 59.80%        | 39.63%           |
| Qwen3-1.7B | Qwen3-4B     | 69.20%         | 73.70%        | 44.51%           |
| Qwen3-1.7B | Qwen3-8B     | 69.15%         | 74.00%        | 40.24%           |
| Qwen3-4B   | — (baseline) | 77.95%         | 73.00%        | 73.17%           |
| Qwen3-4B   | Qwen3-8B     | 77.05%         | 76.80%        | 71.34%           |

## 455 C Full Experimental Results

### 456 C.1 Math Results: Per-Step Performance

457 Table 9 presents per-step results for the primary math experiments (LoRA,  $n=1$ , 3,200 problems).

Table 9: **MATH-500 per-step results.** LoRA,  $n=1$ , bs=16, 3,200 problems. Baseline: 50.95% avg@4.

| Method  | Metric | Step 50      | Step 100     | Step 150     | Step 200     |
|---------|--------|--------------|--------------|--------------|--------------|
| ESR-50  | avg@4  | 62.35        | 66.05        | <b>66.65</b> | 64.85        |
|         | maj@4  | 69.40        | 72.00        | 71.00        | 71.20        |
|         | pass@4 | 77.20        | 79.40        | 81.00        | 79.60        |
| ESR-100 | avg@4  | 63.75        | 64.45        | 65.15        | <b>65.85</b> |
|         | maj@4  | 70.00        | 68.40        | 69.60        | 70.80        |
|         | pass@4 | 79.80        | 78.40        | 80.20        | 79.80        |
| ESR-150 | avg@4  | 65.35        | <b>66.65</b> | 65.30        | 65.75        |
|         | maj@4  | 66.80        | 67.00        | 66.30        | 67.30        |
|         | pass@4 | 79.00        | 81.00        | 78.20        | 80.00        |
| ESR-200 | avg@4  | <b>66.05</b> | 64.65        | 65.10        | 65.55        |
|         | maj@4  | 71.20        | 68.40        | 70.00        | 71.20        |
|         | pass@4 | 81.00        | 79.80        | 80.60        | 80.60        |

### 458 C.2 Math Results: Earlier Series ( $n=16$ , 200 problems)

Table 10: **Complete MATH-500 results, LoRA,  $n=16$ , bs=16.** Best per configuration in **bold**.

| Config                 | Step 50      | Step 100     | Step 150     | Step 200     |
|------------------------|--------------|--------------|--------------|--------------|
| <i>avg@4</i>           |              |              |              |              |
| ESR-5                  | 55.15        | 55.65        | <b>56.50</b> | 56.30        |
| ESR-10                 | <b>59.50</b> | 56.40        | 56.30        | 56.00        |
| ESR-20                 | <b>60.20</b> | 57.35        | 56.40        | 56.55        |
| ESR-50                 | <b>62.45</b> | 62.15        | 60.50        | 61.90        |
| ESR-100                | 61.15        | 63.55        | <b>64.25</b> | 63.85        |
| ESR-150                | 63.55        | <b>65.70</b> | 64.80        | 64.40        |
| ESR-200                | 63.25        | <b>66.75</b> | 64.55        | 65.25        |
| Progressive            | 61.10        | 62.05        | 61.95        | <b>62.30</b> |
| Full-seq <sup>†</sup>  | <b>65.60</b> | 46.30        | 49.70        | 43.80        |
| Full-seq (first-boxed) | <b>65.55</b> | 64.95        | 65.55        | 64.95        |
| <i>pass@4</i>          |              |              |              |              |
| ESR-5                  | 74.0         | 74.8         | <b>76.0</b>  | 73.4         |
| ESR-10                 | <b>75.8</b>  | 73.2         | 75.4         | 74.2         |
| ESR-20                 | <b>77.0</b>  | 76.0         | 73.4         | 75.8         |
| ESR-50                 | 77.0         | 78.4         | 77.6         | <b>78.6</b>  |
| ESR-100                | 77.6         | 79.2         | <b>80.2</b>  | 80.2         |
| ESR-150                | 79.6         | 77.4         | <b>80.0</b>  | 79.4         |
| ESR-200                | 79.6         | <b>80.2</b>  | 79.4         | 79.2         |
| Progressive            | 77.2         | 77.8         | 78.2         | <b>78.8</b>  |
| Full-seq               | <b>80.0</b>  | 72.2         | 77.0         | 74.2         |

<sup>†</sup> Last-boxed extraction; degrades due to repetition.



Table 11: **Complete coding results, LoRA.** HumanEval (HE) and MBPP pass@1.

| Config   | Metric | s50         | s100 | s150        | s200 | s250        | s300 | s350        | s400 |
|----------|--------|-------------|------|-------------|------|-------------|------|-------------|------|
| ESR-50   | HE     | 37.8        | 39.0 | 39.6        | 41.5 | 40.2        | 40.9 | <b>42.1</b> | 40.9 |
|          | MBPP   | 52.1        | 48.9 | 46.0        | 46.0 | 47.6        | 46.3 | 46.6        | 46.3 |
| ESR-100  | HE     | 37.2        | 39.0 | <b>42.1</b> | 37.8 | 39.0        | 37.8 | 37.8        | 38.4 |
|          | MBPP   | 52.1        | 49.7 | 49.2        | 49.2 | 48.4        | 49.2 | 47.9        | 48.9 |
| ESR-150  | HE     | 36.6        | 35.4 | 36.6        | 39.0 | <b>41.5</b> | 39.6 | 38.4        | 37.2 |
|          | MBPP   | 51.3        | 50.0 | 48.4        | 50.5 | 49.5        | 50.3 | 49.7        | 48.7 |
| Full-seq | HE     | <b>40.2</b> | 31.7 | 32.3        | 32.9 | 27.4        | 28.0 | 26.8        | 26.8 |
|          | MBPP   | 52.6        | 48.9 | 49.5        | 48.1 | 47.9        | 47.1 | 48.4        | 47.6 |

Table 12: **Function calling results (BFCL), LoRA.** Name accuracy / Full accuracy / Parse rate. Best full\_acc in **bold**.

| Method               | Best Step  | Name Acc | Full Acc      | Parse Rate |
|----------------------|------------|----------|---------------|------------|
| Baseline             | —          | 9.70%    | 2.70%         | 24.20%     |
| Teacher (Qwen3-1.7B) | —          | 75.30%   | 54.00%        | 75.30%     |
| ESR-50               | 200        | 95.20%   | 57.20%        | 98.30%     |
| <b>ESR-100</b>       | <b>100</b> | 86.20%   | <b>61.30%</b> | 91.30%     |
| ESR-150              | 200        | 88.70%   | 61.50%        | 92.50%     |
| ESR-200              | 200        | 80.80%   | 54.50%        | 90.20%     |
| Full-seq             | 100        | 81.00%   | 58.20%        | 86.70%     |

Table 13: **Scaling results.** All use ESR-100, LoRA, 3,200 problems. Best across training steps. Baseline performance shown for reference.

| Student → Teacher | Math avg@4 | BFCL full_acc | HE pass@1 | MBPP+ pass@1 |
|-------------------|------------|---------------|-----------|--------------|
| M-1.5B baseline   | 50.95%     | 2.70%         | —         | —            |
| M-1.5B → Q3-1.7B  | 65.85%     | 61.30%        | 42.07%    | 44.44%       |
| M-1.5B → Q3-4B    | 68.95%     | 45.80%        | 43.29%    | 47.62%       |
| M-1.5B → Q3-8B    | 67.85%     | 57.00%        | 41.46%    | 44.44%       |
| Q3-1.7B baseline  | 69.20%     | 59.80%        | 39.63%    | 44.71%       |
| Q3-1.7B → Q3-4B   | 69.20%     | 73.70%        | 44.51%    | 47.62%       |
| Q3-1.7B → Q3-8B   | 69.15%     | 74.00%        | 40.24%    | 48.15%       |
| Q3-4B baseline    | 77.95%     | 73.00%        | 73.17%    | 55.56%       |
| Q3-4B → Q3-8B     | 77.05%     | 76.80%        | 71.34%    | 56.35%       |

459 **C.3 Coding Results**460 **C.4 Function Calling Results**461 **C.5 Scaling Experiment Results**462 **C.6 Full-Sequence Degradation Details**

463 Table 14 summarises every full-rollout cell (across pairs, tasks, and regimes) for which we have a  
464 per-step trajectory on disk and observe degradation across training. For each cell we report the *peak*  
465 score (best checkpoint across training) and the *lowest* score (worst checkpoint after the peak), together  
466 with the step indices at which they occur. Bold  $\Delta$  values indicate cells where Full Rollout drops  
467 below the student baseline. The Qwen-math  $n=16$  row (boxed-repetition collapse) is documented in  
468 greater detail in Table 15 below.

469 **Per-step trajectories** for the seven cells in Table 14:

- 470 • **Qwen-1.5B→Q3-1.7B, LoRA, HumanEval (pass@1):** 40.2 (s50) → 31.7 → 32.3 → 32.9  
471 → 27.4 → 28.0 → 26.8 → 26.8 (s400).

Table 14: **Full-rollout collapses across training (peak  $\rightarrow$  lowest).** Each row is a single Full Rollout experiment; we report the highest and lowest checkpoint scores across all saved steps. “ $\Delta$ ” is lowest  $-$  peak. Cells with  $^\dagger$  end the run *below the student baseline*. Trajectories are taken from the per-step eval logs on the training cluster (docs/main\_results.md, docs/scaling\_results.md). ESR on the same cells (not shown here) stays within  $\pm 1$ pp of its peak across training; see Tables 9 and 11.

| Pair                                | Regime  | Task                | Metric | Peak (step) | Lowest (step)           | $\Delta$                   |
|-------------------------------------|---------|---------------------|--------|-------------|-------------------------|----------------------------|
| Qwen-1.5B $\rightarrow$ Q3-1.7B     | LoRA    | MATH-500 ( $n=16$ ) | avg@4  | 65.60 (s50) | 43.80 (s200) $^\dagger$ | <b><math>-21.80</math></b> |
| Qwen-1.5B $\rightarrow$ Q3-4B       | LoRA    | MATH-500            | avg@4  | 67.45 (s50) | 54.45 (s100)            | $-13.00$                   |
| Qwen-1.5B $\rightarrow$ Q3-4B       | LoRA    | MATH-500            | maj@4  | 73.20 (s50) | 63.40 (s200)            | $-9.80$                    |
| Qwen-1.5B $\rightarrow$ Q3-1.7B     | LoRA    | HumanEval           | pass@1 | 40.2 (s50)  | 26.8 (s350)             | $-13.4$                    |
| Qwen-1.5B $\rightarrow$ Q3-1.7B     | LoRA    | HumanEval+          | pass@1 | 35.4 (s50)  | 25.0 (s250)             | $-10.4$                    |
| Qwen-1.5B $\rightarrow$ Q3-1.7B     | FullIFT | HumanEval           | pass@1 | 36.6 (s150) | 30.5 (s300)             | $-6.1$                     |
| Qwen-1.5B $\rightarrow$ Q3-1.7B     | FullIFT | HumanEval+          | pass@1 | 31.7 (s150) | 26.2 (s250)             | $-5.5$                     |
| Gemma-2-2B $\rightarrow$ Gemma-3-4B | LoRA    | HumanEval           | pass@1 | 22.0 (s50)  | 11.6 (s150) $^\dagger$  | <b><math>-10.4</math></b>  |

$^\dagger$  Lowest score is below the student baseline (Qwen-Math-1.5B MATH-500: 50.95%; Gemma-2-2B HumanEval: 23.78%).

- **Qwen-1.5B  $\rightarrow$  Q3-1.7B, LoRA, HumanEval+ (pass@1):** 35.4 (s50)  $\rightarrow$  27.4  $\rightarrow$  29.9  $\rightarrow$  29.9  $\rightarrow$  25.0  $\rightarrow$  26.2  $\rightarrow$  25.0  $\rightarrow$  25.0 (s400).
- **Qwen-1.5B  $\rightarrow$  Q3-1.7B, FullIFT, HumanEval (pass@1):** 32.3  $\rightarrow$  32.3  $\rightarrow$  36.6 (s150)  $\rightarrow$  36.0  $\rightarrow$  31.1  $\rightarrow$  30.5  $\rightarrow$  31.7  $\rightarrow$  31.1 (s400).
- **Qwen-1.5B  $\rightarrow$  Q3-1.7B, FullIFT, HumanEval+ (pass@1):** 27.4  $\rightarrow$  28.7  $\rightarrow$  31.7 (s150)  $\rightarrow$  30.5  $\rightarrow$  26.2  $\rightarrow$  26.2  $\rightarrow$  26.8  $\rightarrow$  26.2 (s400).
- **Gemma-2-2B  $\rightarrow$  Gemma-3-4B, LoRA, HumanEval (pass@1):** 22.0 (s50)  $\rightarrow$  14.6  $\rightarrow$  11.6  $\rightarrow$  11.6 (s200). Falls below the 23.78 student baseline at s100 and stays there.
- **Qwen-1.5B  $\rightarrow$  Q3-4B, LoRA, MATH-500 avg@4:** 67.45 (s50)  $\rightarrow$  54.45  $\rightarrow$  58.85  $\rightarrow$  55.05 (s200) (boxed-repetition pattern; ESR-100 on the same pair climbs to 68.95).
- **Qwen-1.5B  $\rightarrow$  Q3-1.7B, LoRA, MATH-500 ( $n=16$ ) avg@4:** 65.60 (s50)  $\rightarrow$  46.30  $\rightarrow$  49.70  $\rightarrow$  43.80 (s200). Boxed-repetition collapse; statistics in Table 15.

Table 15: **Boxed repetition progression** in full-sequence LoRA distillation on MATH-500 ( $n=16$ , bs=16). Companion to the corresponding row of Table 14.

| Statistic                   | Step 50 | Step 100 | Step 150 | Step 200 |
|-----------------------------|---------|----------|----------|----------|
| Multi-boxed responses       | 2.9%    | 91.5%    | 80.5%    | 89.1%    |
| Avg \boxed{} per response   | 1.0     | 88.1     | 54.1     | 58.0     |
| Repetitive responses        | 5.9%    | 78.8%    | 58.3%    | 77.5%    |
| Avg response length (chars) | 1,388   | 4,023    | 3,585    | 4,558    |
| Official avg@4              | 65.60%  | 46.30%   | 49.70%   | 43.80%   |
| Relaxed avg@4 (GT in text)  | 74.60%  | 70.90%   | 71.90%   | 70.30%   |

## 484 D KL-Profile Diagnostic for Choosing $N$

485 A practical question for a new student–teacher pair is: what value of  $N$  should I use? Our position-  
 486 sweep (Section 4.3) shows a wide plateau on Qwen math ( $N \in [50, 200]$ , avg@4 within  $\sim 1$ pp);  
 487 the same plateau appears on Gemma math ( $N \approx 100$ ). These plateau locations coincide with the  
 488 position at which the cumulative per-position KL  $\sum_{k \leq N} \mathbb{E}[\text{KL}_k]$  reaches  $\sim 45\%$  of the full-trajectory  
 489 total, measured on a small ( $\sim 1$ k trajectory) pre-training rollout. This suggests a cheap rule of thumb:  
 490 run a 1k-trajectory per-position KL scan before training, and pick  $N$  at the 45% cumulative-KL  
 491 point. We verify this on two cells (Qwen, Gemma math) and leave a full six-cell verification ( $\{\text{Qwen},$   
 492  $\text{Gemma}\} \times \{\text{math}, \text{coding}, \text{funcall}\}$ ) as a short follow-up (**TBD: KL-profile scan**); the diagnostic is  
 493 reported here as a practical recipe rather than a claim.

## E Token Classification Methodology

We classify each token into six categories based on string matching:

1. **planning**: Reasoning keywords (“To”, “Let”, “First”, “Step”, “We”, “Given”, “Therefore”, “Thus”, “Since”).
2. **structural**: Punctuation, whitespace, formatting tokens.
3. **math\_number**: Digits (0–9).
4. **math\_operator**: Arithmetic operators (+, −, ×, /, =).
5. **math\_latex**: LaTeX delimiters (\(, \[).
6. **continuation**: All others.

Table 16: Mean KL by token category and position range.

| Category      | 0–4  | 5–19 | 20–49 | 50–99 | 100–199 | 200–499 |
|---------------|------|------|-------|-------|---------|---------|
| planning      | 4.50 | 0.79 | 1.49  | 1.66  | 1.49    | 2.37    |
| structural    | 3.26 | 1.46 | 1.60  | 0.93  | 0.60    | 0.86    |
| math_number   | 1.49 | 0.60 | 0.74  | 0.28  | 0.17    | 0.13    |
| math_operator | 7.30 | 1.84 | 0.81  | 0.37  | 0.21    | 0.14    |
| math_latex    | 8.84 | 9.23 | 6.50  | 4.95  | 2.97    | 1.87    |
| continuation  | 1.94 | 1.12 | 1.19  | 0.89  | 0.70    | 0.48    |

Table 17: Top 20 highest-KL tokens (minimum 50 occurrences across 10,000 trajectories).

| Rank | Token           | Category     | Count  | Mean KL |
|------|-----------------|--------------|--------|---------|
| 1    | “Solution”      | planning     | 152    | 21.93   |
| 2    | “Analysis”      | continuation | 125    | 16.49   |
| 3    | \[              | math_latex   | 7,152  | 13.21   |
| 4    | “examines”      | continuation | 74     | 11.51   |
| 5    | “He”            | continuation | 150    | 10.80   |
| 6    | \(              | math_latex   | 21,243 | 10.30   |
| 7    | “First”         | planning     | 1,706  | 9.98    |
| 8    | “tests”         | continuation | 52     | 8.71    |
| 9    | \%              | math_latex   | 82     | 8.68    |
| 10   | “There”         | continuation | 201    | 8.28    |
| 11   | “Therefore”     | planning     | 4,913  | 7.95    |
| 12   | “Identify”      | continuation | 1,345  | 7.78    |
| 13   | “To”            | planning     | 8,806  | 6.90    |
| 14   | “When”          | continuation | 89     | 6.62    |
| 15   | “This”          | continuation | 621    | 6.34    |
| 16   | “Thus”          | planning     | 2,547  | 6.24    |
| 17   | “First” (space) | planning     | 259    | 6.10    |
| 18   | “To” (space)    | planning     | 1,174  | 5.37    |
| 19   | “The”           | planning     | 2,626  | 5.25    |
| 20   | “Next”          | planning     | 1,753  | 5.03    |

## NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

**The checklist answers are an integral part of your paper submission.** They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive” or “we were unable to find the license for the dataset we used”). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- Delete this instruction block, but keep the section heading “NeurIPS Paper Checklist”.
- Keep the checklist subsection headings, questions/answers and guidelines below.
- Do not modify the questions and only use the provided macros for your answers.

### 1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

### 2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

### 3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

### 4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not include experiments.

- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

## 5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

## 6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

## 7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The authors should answer **[Yes]** if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

## 8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer **[N/A]** means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.



- 707           • The paper should disclose whether the full research project required more compute  
708           than the experiments reported in the paper (e.g., preliminary or failed experiments that  
709           didn't make it into the paper).

710   **9. Code of ethics**

711   Question: Does the research conducted in the paper conform, in every respect, with the  
712   NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines?>

713   Answer: **[TODO]**

714   Justification: **[TODO]**

715   Guidelines:

- 716           • The answer [N/A] means that the authors have not reviewed the NeurIPS Code of  
717           Ethics.  
718           • If the authors answer [No], they should explain the special circumstances that require a  
719           deviation from the Code of Ethics.  
720           • The authors should make sure to preserve anonymity (e.g., if there is a special consid-  
721           eration due to laws or regulations in their jurisdiction).

722   **10. Broader impacts**

723   Question: Does the paper discuss both potential positive societal impacts and negative  
724   societal impacts of the work performed?

725   Answer: **[TODO]**

726   Justification: **[TODO]**

727   Guidelines:

- 728           • The answer [N/A] means that there is no societal impact of the work performed.  
729           • If the authors answer [N/A] or [No], they should explain why their work has no societal  
730           impact or why the paper does not address societal impact.  
731           • Examples of negative societal impacts include potential malicious or unintended uses  
732           (e.g., disinformation, generating fake profiles, surveillance), fairness considerations  
733           (e.g., deployment of technologies that could make decisions that unfairly impact specific  
734           groups), privacy considerations, and security considerations.  
735           • The conference expects that many papers will be foundational research and not tied  
736           to particular applications, let alone deployments. However, if there is a direct path to  
737           any negative applications, the authors should point it out. For example, it is legitimate  
738           to point out that an improvement in the quality of generative models could be used to  
739           generate Deepfakes for disinformation. On the other hand, it is not needed to point out  
740           that a generic algorithm for optimizing neural networks could enable people to train  
741           models that generate Deepfakes faster.  
742           • The authors should consider possible harms that could arise when the technology is  
743           being used as intended and functioning correctly, harms that could arise when the  
744           technology is being used as intended but gives incorrect results, and harms following  
745           from (intentional or unintentional) misuse of the technology.  
746           • If there are negative societal impacts, the authors could also discuss possible mitigation  
747           strategies (e.g., gated release of models, providing defenses in addition to attacks,  
748           mechanisms for monitoring misuse, mechanisms to monitor how a system learns from  
749           feedback over time, improving the efficiency and accessibility of ML).

750   **11. Safeguards**

751   Question: Does the paper describe safeguards that have been put in place for responsible  
752   release of data or models that have a high risk for misuse (e.g., pre-trained language models,  
753   image generators, or scraped datasets)?

754   Answer: **[TODO]**

755   Justification: **[TODO]**

756   Guidelines:

- 757           • The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

## 12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, [paperswithcode.com/datasets](https://paperswithcode.com/datasets) has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

## 13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

## 14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: **[TODO]**

Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

#### 15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

#### 16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [TODO]

Justification: [TODO]

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.